# Type MS · The TCP/IP model: four layers and what each does

## Answer 1 of 6 · Question 3 part b · 4 marks

2021 NOV · Paper 01 · extract 1 of 5 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 3(a) | Award **one** mark for correct network type<br>Award **one** mark for correct usage model<br>One mark for each column | | 2 |

| Network type | Tick (□) | Usage model | Tick (□) |
|---|---|---|---|
| Local Area Network (LAN) | | Client-Server | |
| Wide Area Network (WAN) | | Peer-to-peer | □ |
| Personal Area Network (PAN) | □ | | |

# Type MS · The TCP/IP model: four layers and what each does

## Answer 1 of 6 · Question 3 part b · 4 marks

2021 NOV · Paper 01 · extract 2 of 5 · find part b in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)</td><td>Award one mark for each correct layer name up to a maximum of two<br>Award one mark per layer for a correct function associated with the correct layer up to a maximum of two<br><br><table><tr><th>Layer</th><th>Function</th></tr><tr><td>Application</td><td>• Selects and uses the correct protocol to transmit data<br>• Interacts with the user</td></tr><tr><td>Transport</td><td>• Identifies the server port to use<br>• Identifies the client port to use<br>• Divides the data into packets<br>• Numbers the packets<br>• Adds the total number of packets<br>• Sets up communication between hosts / Establishes end to end communication<br>• Passes the packets to the network layer<br>• Checks the packets arrive at the destination<br>• Resends any packets that have not arrived</td></tr><tr><td>Network</td><td>• Adds the source/sender's IP address<br>• Adds the destination IP address<br>• Routes the packets</td></tr><tr><td>Data link</td><td>• Controls physical connections between pieces of hardware<br>• Adds MAC addresses to the packets<br>• Sends the packets on their way<br>• Adds headers and trailers</td></tr></table></td><td>Accept functions at the receiving end too</td><td>4</td></tr>
</table>

# Type MS · The TCP/IP model: four layers and what each does

## Answer 1 of 6 · Question 3 part b · 4 marks

2021 NOV · Paper 01 · extract 3 of 5 · find part b in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(c) | **The only correct answer is A**<br><br>B *is not correct because* it is not a smartphone frequency band<br>C *is not correct because* it is not a smartphone frequency band<br>D *is not correct because* it is not a smartphone frequency band | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(d)(i) | Award **one** mark from:<br><br>• higher frequency has more waves per second/carries more data in the same time (1)<br>• higher frequency gives greater bandwidth (1)<br>• higher frequency is more stable/less prone to interference (1) | | 1 |
| 3(d)(ii) | Award up to **two** marks for a linked explanation such as:<br><br>• gives Carlo faster responses to his communications/connections (1) allowing Carlo to move/download/share data in a shorter time (1)<br>• gives Carlo more reliable communications/can access communications more widely/in more places (1) giving Carlo a better user experience (1)<br>• gives Carlo more secure communications (1) improving his privacy/reducing other people's ability to intercept/spy on his communications (1) | | 2 |

# Type MS · The TCP/IP model: four layers and what each does

## Answer 1 of 6 · Question 3 part b · 4 marks

2021 NOV · Paper 01 · extract 4 of 5 · find part b in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(e)(i) | Award **one** mark from:<br><br>• do not leave the smartphone on standby/turn off when not being used (1)<br>• use (portable) solar cells/other renewable generation method (1)<br>• unplug charger when not in use (1)<br>• reduce screen brightness (1)<br>• turn off WiFi/GPS/Bluetooth/location (when not in use) (1)<br>• close apps that are not being used (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 3(e)(ii) | Award up to **two** marks for a linked explanation.<br><br>Award **one** mark for the action<br>• keep smartphone for longer (1)<br>• repair/recycle smartphone if not working (1)<br>• upgrade the operating system (1)<br><br>Award **one** mark for the impact<br>• less e-waste is generated (1)<br>• manufacturing could be reduced (1) | | 2 |
| | | **Total** | **13** |

# Type MS · The TCP/IP model: four layers and what each does

## Answer 1 of 6 · Question 3 part b · 4 marks

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| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · The TCP/IP model: four layers and what each does

Answer 2 of 6 · Question 1 part f.ii · 2 marks

2020 NOV · Paper 01 · extract 1 of 5 · find part f.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(a) | Any **two** from:<br>• (Share) access to the Internet/WWW/broadband connection (1)<br>• (Internal) communication (using email/instant messaging/calendars) (1)<br>• Share files/data (1)<br>• Share peripherals/printers / hardware (1)<br>• Increases the amount of storage (since network storage devices can be attached) (1)<br>• Saves money on licences (since network site licences for software are usually cheaper than buying a stand-alone licence for every machine) (1)<br>• Centralised backup (1)<br>• Centralised security (1) | • Award examples if mapped to a bullet point<br>• Do not award the same bullet more than once<br>• Do not award for install updates (in the stem) | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b) (i) | Any **two** from:<br>• ~~Faster (data transfer) (1)~~ Not really true, its Greater bandwidth / more bits per second (1)<br>• The connection does not get worse the further you are from the router / more reliable (1)<br>• Connection does not get obstructed by walls, ceilings, and furniture (1)<br>• More secure (1) | Accept reverse arguments for disadvantages of wireless. | 2 |

Type MS · The TCP/IP model: four layers and what each does

Answer 2 of 6 · Question 1 part f.ii · 2 marks

2020 NOV · Paper 01 · extract 2 of 5 · find part f.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b) (ii) | Any **two** from:<br>• Installation/maintenance is more complex (1)<br>• Devices need to be physically connected (1)<br>• Less portable / limited by length of cable (1)<br>• Limited number of devices can be connected (1)<br>• Some digital devices cannot use a wired connection (1)<br>• Trip hazard (1) | Accept reverse arguments for advantages of wireless. | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(c) | Award **two** marks for a linked description that addresses the individual mark points:<br><br>The internet is a (global) network of networks/ connected devices (1) whereas the www is the collection of web pages/service accessed using the internet (1) / the www is resources located via URLs/domain names (1)<br><br>The internet is the infrastructure (1) and the www is a service that runs on that infrastructure (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|



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Type MS · The TCP/IP model: four layers and what each does

Answer 2 of 6 · Question 1 part f.ii · 2 marks

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| | | | |
|---|---|---|---|
| 1(d) | Award **two** marks for a linked explanation such as:<br><br>Running out of IPv4 addresses/the number of (internet) connected devices has grown (1)<br><br>IPv6 addresses are long/longer than IPv4 / the number of possible addresses is large/will last much longer (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e)(i) | Star | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e) (ii) | Any **one** from:<br><br>• easy to connect/remove new nodes (1)<br>• failure of one node/link does not affect the rest of the network (1)<br>• easy to detect the failure of one node/link (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e) (iii) | Any **one** from:<br><br>• if central switch/hub fails then the whole network fails (1)<br>• performance and number of nodes that can be added depends on capacity of central switch/hub (1)<br>• can require more cable than some of the other topologies (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|



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Type MS · The TCP/IP model: four layers and what each does

Answer 2 of 6 · Question 1 part f.ii · 2 marks

2020 NOV · Paper 01 · extract 4 of 5 · find part f.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 1(f)(i) | Any **two** from:<br>• SMTP (1)<br>• IMAP (1)<br>• POP/POP3 (1) | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(f)(ii) | Any **two** from:<br>• passes the (reassembled) packets to the application layer (1)<br>• check if all packets have arrived (1)<br>• determine whether the contents are correct (1)<br>• requests resending of lost or damaged packets (1)<br>• reassembles packets in correct order/ into an email (1) | Do not accept receives from network layer (stem) | 2 |



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## Type MS · The TCP/IP model: four layers and what each does

### Answer 2 of 6 · Question 1 part f.ii · 2 marks

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| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · The TCP/IP model: four layers and what each does

Answer 3 of 6 · Question 2 part c.i · 1 mark

2021 JUN · Paper 01 · extract 1 of 5 · find part c.i in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 2(a) | Award **one** mark from:<br><br>• share peripherals/devices (e.g. printers, scanners) (1)<br>• communicate (e.g. email, instant messaging, play games) (1)<br>• share data (e.g. files/music/videos/backups on servers) (1)<br>• deploy/update applications/software(1)<br>• administrative purposes (e.g. remote access/desktop) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(b) | Award **one** mark from:<br><br>• to allow (connected) machines to communicate (1)<br>• to provide the rules of communication between (networked) devices (1) | Accept any other response indicating protocols enable communication between devices | 1 |

## Type MS · The TCP/IP model: four layers and what each does

### Answer 3 of 6 · Question 2 part c.i · 1 mark

2021 JUN · Paper 01 · extract 2 of 5 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c)(i) | **The only correct answer is A**<br><br>***B*** *is not correct because FTP is an application protocol, which is added in the application layer*<br><br>***C*** *is not correct because FTP is an application protocol, which is added in the application layer*<br><br>***D*** *is not correct because FTP is an application protocol, which is added in the application layer* | | 1 |
| 2(c)(ii) | Award **one** mark from:<br><br>• Adds the source/destination IP address (1)<br>• Establishes sockets (an IP address followed by a port number) (1)<br>• Moves packets onto the next network node (1)<br>• Strips the source/destination IP addresses (when it arrives at its destination) (1)<br>• Passes the packet to the transport/next layer (when it arrives at its destination) (1)<br>• Adds/removes packet headers (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(d) | Award **one** mark for each of:<br><br>**Domain Name** / **IP Address** / **URL**<br>(blank) / (blank) / ✓<br>(blank) / ✓ / (blank)<br>✓ / (blank) / (blank) | | 3 |

# Type MS · The TCP/IP model: four layers and what each does

## Answer 3 of 6 · Question 2 part c.i · 1 mark

2021 JUN · Paper 01 · extract 3 of 5 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(e)(i) | Bus | | 1 |
| 2(e)(ii) | Award **one** mark from:<br>• Easy to connect/remove devices/computers (in a network) (1)<br>• Cable cost less / fewer cables needed (compared to mesh/star topology) (1)<br>• Easier to plan/setup/maintain (1) | | 1 |
| 2(e)(iii) | Award **one** mark from:<br>• If the main cable/a terminator fails or gets damaged the whole network will fail (1)<br>• The more workstations the slower it gets (1)<br>• The more workstations the more collisions (1)<br>• Every device 'sees' all of the data on the network (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(f)(i) | Award **two** marks from:<br>• Reduces time taken to upload/download (1)<br>• Reduces the bandwidth required/cost of data to upload/download (1)<br>• (Usually) produces smaller file/reduces storage (on the web server/device) (1)<br>• People have a limited hearing range / can't tell the difference (1)<br>• People's brains will 'fill in the blanks' (1) | | 2 |
| 2(f)(ii) | **The only correct answer is C**<br><br>***A*** *is not correct because it is not compressed*<br><br>***B*** *is not correct because it uses lossless compression*<br><br>***D*** *is not correct because it uses lossless compression* | | 1 |
| 2(f)(iii) | Award **one** mark from:<br>• 3c3w2c6w3c (1)<br>• c3w3c2w6c3 (1) | | 1 |

Type MS · The TCP/IP model: four layers and what each does

Answer 3 of 6 · Question 2 part c.i · 1 mark

2021 JUN · Paper 01 · extract 4 of 5 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(g) | Award **four** marks for a correct expression or correct answer<br><br>**Examples of correct expressions:**<br>6 x 1000 x 1000 x 1000 x 8 (4)<br>54 x 1000 x 1000<br><br>1000 x 8 (4)<br>9<br><br>888.9 (4)<br><br>For an incorrect or partial expression award **one** mark for each of:<br><br>• **Bits to transfer**<br>○ $6 \times 1000^3$ (1)<br>○ x 8 (or equivalent) (1)<br><br>• **Speed in bits per second**<br>○ $54 \times 1000^2$ (1)<br><br>• Their bits divided by their speed (1) | • Units are not required<br>• Equivalent expressions are accepted<br>• Allow follow through for incorrect or partial expressions | 4 |

## Type MS · The TCP/IP model: four layers and what each does

### Answer 3 of 6 · Question 2 part c.i · 1 mark

2021 JUN · Paper 01 · extract 5 of 5 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · The TCP/IP model: four layers and what each does

Answer 4 of 6 · Question 2 part c.ii · 1 mark

2021 JUN · Paper 01 · extract 1 of 5 · find part c.ii in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 2(a) | Award **one** mark from:<br><br>• share peripherals/devices (e.g. printers, scanners) (1)<br>• communicate (e.g. email, instant messaging, play games) (1)<br>• share data (e.g. files/music/videos/backups on servers) (1)<br>• deploy/update applications/software(1)<br>• administrative purposes (e.g. remote access/desktop) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(b) | Award **one** mark from:<br><br>• to allow (connected) machines to communicate (1)<br>• to provide the rules of communication between (networked) devices (1) | Accept any other response indicating protocols enable communication between devices | 1 |

Type MS · The TCP/IP model: four layers and what each does

Answer 4 of 6 · Question 2 part c.ii · 1 mark

2021 JUN · Paper 01 · extract 2 of 5 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c)(i) | **The only correct answer is A**<br><br>***B*** *is not correct because FTP is an application protocol, which is added in the application layer*<br><br>***C*** *is not correct because FTP is an application protocol, which is added in the application layer*<br><br>***D*** *is not correct because FTP is an application protocol, which is added in the application layer* | | 1 |
| 2(c)(ii) | Award **one** mark from:<br><br>• Adds the source/destination IP address (1)<br>• Establishes sockets (an IP address followed by a port number) (1)<br>• Moves packets onto the next network node (1)<br>• Strips the source/destination IP addresses (when it arrives at its destination) (1)<br>• Passes the packet to the transport/next layer (when it arrives at its destination) (1)<br>• Adds/removes packet headers (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(d) | Award **one** mark for each of:<br><br>**Domain Name** / **IP Address** / **URL**<br>(blank) / (blank) / ✓<br>(blank) / ✓ / (blank)<br>✓ / (blank) / (blank) | | 3 |

Type MS · The TCP/IP model: four layers and what each does

Answer 4 of 6 · Question 2 part c.ii · 1 mark

2021 JUN · Paper 01 · extract 3 of 5 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(e)(i) | Bus | | 1 |
| 2(e)(ii) | Award **one** mark from:<br>• Easy to connect/remove devices/computers (in a network) (1)<br>• Cable cost less / fewer cables needed (compared to mesh/star topology) (1)<br>• Easier to plan/setup/maintain (1) | | 1 |
| 2(e)(iii) | Award **one** mark from:<br>• If the main cable/a terminator fails or gets damaged the whole network will fail (1)<br>• The more workstations the slower it gets (1)<br>• The more workstations the more collisions (1)<br>• Every device 'sees' all of the data on the network (1) | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(f)(i) | Award **two** marks from:<br>• Reduces time taken to upload/download (1)<br>• Reduces the bandwidth required/cost of data to upload/download (1)<br>• (Usually) produces smaller file/reduces storage (on the web server/device) (1)<br>• People have a limited hearing range / can't tell the difference (1)<br>• People's brains will 'fill in the blanks' (1) | | 2 |
| 2(f)(ii) | **The only correct answer is C**<br>***A** is not correct because it is not compressed*<br>***B** is not correct because it uses lossless compression*<br>***D** is not correct because it uses lossless compression* | | 1 |
| 2(f)(iii) | Award **one** mark from:<br>• 3c3w2c6w3c (1)<br>• c3w3c2w6c3 (1) | | 1 |

Type MS · The TCP/IP model: four layers and what each does

Answer 4 of 6 · Question 2 part c.ii · 1 mark

2021 JUN · Paper 01 · extract 4 of 5 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(g) | Award **four** marks for a correct expression or correct answer<br><br>**Examples of correct expressions:**<br>6 x 1000 x 1000 x 1000 x 8 (4)<br>54 x 1000 x 1000<br><br>1000 x 8 (4)<br>9<br><br>888.9 (4)<br><br>For an incorrect or partial expression award **one** mark for each of:<br><br>• **Bits to transfer**<br>○ $6 \times 1000^3$ (1)<br>○ x 8 (or equivalent) (1)<br><br>• **Speed in bits per second**<br>○ $54 \times 1000^2$ (1)<br><br>• Their bits divided by their speed (1) | • Units are not required<br>• Equivalent expressions are accepted<br>• Allow follow through for incorrect or partial expressions | 4 |

## Type MS · The TCP/IP model: four layers and what each does

### Answer 4 of 6 · Question 2 part c.ii · 1 mark

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| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · The TCP/IP model: four layers and what each does

### Answer 5 of 6 · Question 1 part d · 1 mark

2024 JUN · Paper 01 · extract 1 of 3 · find part d in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 1(a) | Award **one** mark for any of the following up to a maximum of **two** marks:<br>• WANs cover large/wide (geographic) distances/areas / LANs cover small/local (geographic) distances/areas (1)<br>• WANs have lower bandwidth / LANs have higher bandwidth (1)<br>• WANs connect multiple LANs / LANs are single networks (1)<br>• WANs have higher latency / LANs have lower latency (1)<br>• WANs use shared infrastructure / LANs are usually privately owned (1)<br>• WANs use third party connections / LANs are owned by the organisation (1)<br>• WANs have more users/devices / LANs have fewer users (1) | Do not award 2 marks for the same point e.g. A WAN is used over a large geographic area whereas a LAN covers a small distance.<br><br>Two different differences must be identified.<br><br>Do not accept general points such as WANs are slower - too vague.<br><br>Do not accept LANs can be wireless. | 2 |
| 1(b) | Award **one** mark for any of the following:<br>• (Increased risk of) malware/viruses **spreading** / Pharming / Cyber attack (1)<br>• (Increased risk of) hacking/unauthorised access/interception (1) | | 1 |
| 1(c)(i) | Award **one** mark for:<br>• (Partially connected) mesh (1) | | 1 |

## Type MS · The TCP/IP model: four layers and what each does

### Answer 5 of 6 · Question 1 part d · 1 mark

2024 JUN · Paper 01 · extract 2 of 3 · find part d in this mark scheme extract

| | | | |
|---|---|---|---|
| 1c(ii) | Award **one** mark for any of the following:<br>• More expensive to install (1)<br>• Needs more cabling/wiring (1)<br>• Requires more maintenance (1)<br>• Harder to maintain / manage / install (1)<br>• Higher power consumption (1) | | 1 |
| 1(c)(iii) | Award up to **two** marks for a linked explanation such as:<br>• In the event of a damaged/congested connection data/packets can still be routed to their destination (1) because each node/computer is connected to at least two other nodes / there are multiple pathways available (1) | | 2 |
| 1(d) | **The only correct answer is D**<br>*A is not correct because the application layer deals with applications*<br>*B is not correct because the transport layer deals with packet structure*<br>*C is not correct because the network layer deals with routing end to end* | | 1 |
| 1(e)(i) | Award **one** mark for:<br>• https/hypertext transfer protocol secure (1) | | 1 |
| 1(e)(ii) | Award **one** mark for:<br>• specification(.html) (1) | | 1 |
| | | **Total for question 1** | **10** |

## Type MS · The TCP/IP model: four layers and what each does

### Answer 5 of 6 · Question 1 part d · 1 mark

2024 JUN · Paper 01 · extract 3 of 3 · find part d in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

# Type MS · The TCP/IP model: four layers and what each does

## Answer 6 of 6 · Question 1 part b · 2 marks

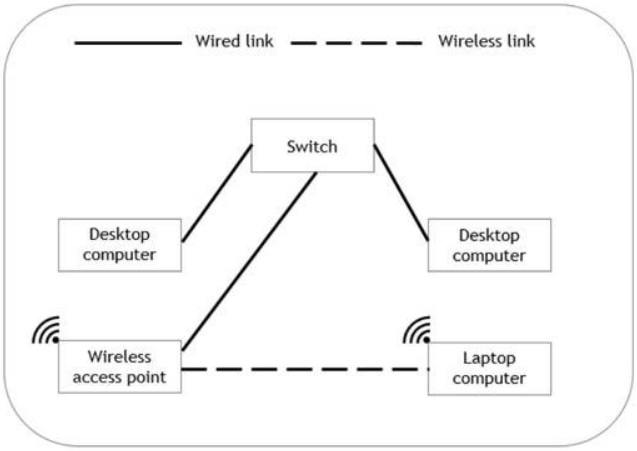
2025 JUN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

<table>
<tr><td>1(a)</td><td>Award one mark for each correct row:<br><br><table><tr><th>Statement</th><th>DNS</th><th>PAN</th><th>WAN</th><th>WWW</th></tr><tr><td>A network of networks not limited to a single location</td><td></td><td></td><td>✓</td><td></td></tr><tr><td>A network that connects devices within the range of a single user</td><td></td><td>✓</td><td></td><td></td></tr><tr><td>A set of hypertext documents accessed through the HTTP protocol</td><td></td><td></td><td></td><td>✓</td></tr><tr><td>A system that maps domain names to IP addresses</td><td>✓</td><td></td><td></td><td></td></tr></table></td><td>Do not award a mark for a row containing more than one tick.</td><td>4</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(b)</td><td>Award one mark for each of:<br><br>• Transport (layer) (1)<br>• (Data) link / network link / network interface (layer) (1)</td><td>Accept<br>• transfer or transmission for Transport<br><br>Do not accept:<br>• Network layer and Application layer (in the question)<br>• Data access layer, Internet layer (alternative names for network layer)<br>• Data’ or ‘network’ without ‘link’ or ‘interface’</td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(c)</td><td>The only correct answer is D<br><br>A is not correct because HTTP is used to transfer hypertext<br>B is not correct because it is used to view email held on a server<br>C is not correct because it is used to download email from a server</td><td></td><td>1</td></tr>
</table>

## Type MS · The TCP/IP model: four layers and what each does

### Answer 6 of 6 · Question 1 part b · 2 marks

2025 JUN · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(d) | Award **one** mark for each correct connection:<br>• One solid line from wireless access point to switch (1)<br>• One solid line from each desktop to switch (1)<br>• One dashed line from laptop computer to wireless access point (1)<br>  | No marks should be awarded if any additional lines have been drawn for each mark point.<br>Do not double penalise (e.g. if you have penalised a second line from desktop to WAP then don't penalise extra line from WAP to desktop) | 3 |

## Type MS · The TCP/IP model: four layers and what each does

### Answer 6 of 6 · Question 1 part b · 2 marks

2025 JUN · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e) | Award up to **two** marks for a linked description such as:<br>• To find weaknesses in a system (1) fix/repair weakness (1)<br>• To find vulnerabilities in a system (1) by testing the effectiveness of the security measures in place (1)<br>• An ethical hacking attack on a network (1) to identify vulnerabilities / recommend measures/take action to improve security (1)<br>• Tests whether users of the network (1) can be tricked into giving away information / giving criminals access to the network / downloading malware (1)<br>• To test the system security (1) / sanctioned by the owner (1) | Explanation can be a combination of different marking points. E.g. MP1a and MP2b | 2 |
| | | **Total for question 1** | **12** |



Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

## Type MS · The TCP/IP model: four layers and what each does

### Answer 6 of 6 · Question 1 part b · 2 marks

2025 JUN · Paper 01 · extract 4 of 4 · find part b in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|